DIPLOMA PROGRAMME PRACTICE · SL · SET B

Chemistry / Paper 2

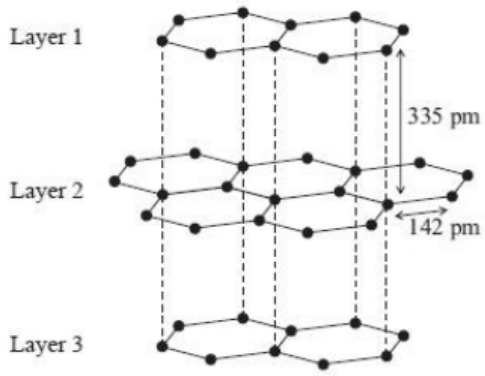
Question booklet · 50 marks · 90 minutes

Answer every part. Show working, units and suitable precision. Use a calculator and a clean Chemistry data booklet. Use separate answer paper for additional working and diagrams.

QUESTION 1 [10]

Layered carbon in an electrolytic cell

A graphite electrode is made from the material shown. The short in-layer C–C distance is 142 pm and the separation between layers is 335 pm. The electrode is then used in molten $PbBr_2$ with inert electrodes.



Stacked sheets of carbon atoms arranged in hexagons within each graphite layer.

(a) Compare the two distances numerically and explain what the difference indicates about the interactions. **[3]**

(b) Explain how graphite can conduct parallel to its layers and why the layers can slide. **[2]**

(c) Write both electrode equations for molten $PbBr_2$ and identify the electrode where oxidation occurs. **[3]**

Paper 2 · Question 1 (continued)

(d) Determine the mass of lead when 0.0250 mol Br_2 is collected under conditions of complete product recovery. Use M(Pb)=207.2 g mol^{-1}. **[2]**

QUESTION 2 [10]

Analysing a gas blend

A 100.0 cm³ mixture contains only CH_4 and C_2H_6. Complete combustion in excess oxygen gives 145.0 cm³ of dry CO_2. All gas volumes are compared at the same temperature and pressure; water is removed before measuring CO_2.

(a) Write balanced equations for complete combustion of each hydrocarbon. **[2]**

(b) Determine the volume percentage of each hydrocarbon. **[3]**

(c) Calculate the oxygen volume consumed and the volume remaining if the original oxygen supply was 300.0 cm³. **[3]**

(d) A leak removes some CO_2 before its measurement. Explain the error in the inferred ethane percentage and one way to check the collection apparatus. **[2]**

QUESTION 3 [10]

Ammonia and a strong acid

A flask contains 40.0 cm³ of 0.150 mol dm⁻³ NH_3(aq). It receives 30.0 cm³ of 0.250 mol dm⁻³ HCl. Assume acid transfer to NH_3 is quantitative, volumes are additive, and remaining strong acid controls final pH.

(a) Write the proton-transfer equation using H_3O^+ and identify the two conjugate acid–base pairs. **[3]**

(b) Determine the limiting reactant and the amount of excess strong acid. **[3]**

(c) Determine the final pH. Explain why simply averaging the initial pH values would fail. **[3]**

(d) Explain the origin of the new N–H bonding pair when NH_3 accepts a proton. **[1]**

QUESTION 4 [10]

Reversing a dissolution measurement

A 5.00 g salt sample, molar mass 80.0 g mol^{-1}, dissolves in 95.0 g of water in a cup. The temperature falls from 23.40 to 20.10 °C. Approximate solution c as 4.18 J g^{-1} K^{-1} and neglect cup heat capacity.

(a) Determine the heat change of the solution and the molar enthalpy of dissolution. **[3]**

(b) Explain the sign using the competing energy changes when an ionic solid dissolves. **[2]**

(c) Each temperature reading has maximum uncertainty ±0.05 °C. With other quantities exact, estimate the maximum percentage and absolute uncertainties in ΔH. **[3]**

(d) If heat enters from the warmer surroundings during dissolution, explain how the reported ΔH changes. **[2]**

QUESTION 5 [10]

Following a four-carbon alcohol

A compound with formula $C_4H_{10}O$ contains one hydroxy group. On controlled oxidation it gives a ketone. Compare straight-chain alcohols only.

(a) Identify the alcohol and the ketone by name and condensed structural formula. **[3]**

(b) Write the oxidation equation using [O] and show the change in oxidation state of the carbon bearing OH. **[3]**

(c) Give the names and condensed formulas of the two successive oxidation products of its straight-chain primary alcohol isomer. **[2]**

(d) Explain why butan-2-ol has a higher normal boiling temperature than an ether of similar molar mass that lacks an O–H bond. **[2]**